

An Impedance String in Physical Time: Complete Spectrum and Transparent Extinction

HCS-C396 theorem and reproducibility package

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Abstract

We reconstruct the full energy-space dynamics of a homogeneous string with one fixed endpoint and one finite nonnegative impedance. A unitary characteristic unfolding preserves physical time and transforms the wave generator into a derivative with one scalar boundary condition. For every nonzero signed round-trip coefficient, the entire spectrum is a simple bilateral arithmetic ladder with a complete Riesz basis. The semigroup norm is an exact staircase, including all round-trip endpoints. At transparent impedance the generator spectrum is empty, yet the semigroup has norm one until precisely twice the length divided by the wave speed; after that time it vanishes. An explicit all-plane resolvent verifies that empty spectrum is an operator theorem, not a failed eigenvalue search. Reciprocal positive impedances give the same norm and different spectral ladders. The classical absorbing-wave mechanism retains its literature ownership. This draft closes the domain, clock, complete spectrum and all-time law, while a finite exact and high-precision ledger audits their constants.

Keywords: impedance string; physical-time semigroup; empty spectrum; exact pseudospectrum; Volterra resolvent; determinant boundary.

中文摘要：

本文完整整理一端固定、另一端具有有限非负阻抗的均匀弦动力学。

精确等距展开保留物理时间，将波动生成元化为带单个边界条件的一阶导数。

非零往返反射系数给出全部简单谱点及完备基，半群范数呈精确阶梯。

透明边界下生成元没有谱点，但演化在完整往返时间以前仍具有单位范数，

此后才完全消亡。全文明确区分无限维证明与有限回归检查，不宣称经典公式的首创。

关键词：阻抗弦；物理时间；空谱；精确拟谱；预解式；行列式边界。

1 One source, three different spectral questions

Perfect absorption separates three quantities that are often conflated: the spectrum of an unbounded generator, the norm of its evolution, and the norm of its resolvent. Our source is the homogeneous wave equation

$$u_{tt} = c^2 u_{xx}, \quad 0 < x < L, \quad u(0, t) = 0, \quad u_x(L, t) + (\eta/c)u_t(L, t) = 0, \quad (1)$$

where $L, c > 0$ and $0 \leq \eta < \infty$. The time in this equation is never replaced by a spectral or target-dependent clock. Define

$$\ell = L/c, \quad \tau = 2L/c, \quad q = (\eta - 1)/(\eta + 1) \in [-1, 1]. \quad (2)$$

The signed coefficient q includes the reflection at the fixed endpoint; it is the negative of the right-end reflection coefficient alone.

Driscoll and Trefethen [1] established the classical absorbing-wave mechanism, the exact decay staircase and the empty-spectrum transparent case. Their Theorem 1 gives resolvent bounds and invariances. We retain that ownership. The package contribution is a self-contained domain-and-clock reconstruction joined to independently executable evidence and explicit operator-category limitations, not a claim of literature priority. The source theorem is substantially different from free-wave observability: there is no boundary control input, observation inequality or control cost.

2 The physical domain and its unitary unfolding

Put $v = u_t$ and $p = cu_x$. On the complex Hilbert space

$$\mathcal{H} = \{u \in H^1(0, L) : u(0) = 0\} \times L^2(0, L), \quad \|(u, v)\|_E^2 = \int_0^L (|p|^2 + |v|^2) dx, \quad (3)$$

the generator is $G_\eta(u, v) = (v, c^2 u_{xx})$. Its domain consists exactly of $u \in H^2, v \in H^1$, with $u(0) = v(0) = 0$ and the right boundary in (1). Physical energy is one half of the squared norm.

Theorem 1 (Exact domain equivalence). *The following map is unitary from $\mathcal{H}$ to $L^2(0, \tau)$:*

$$U(u, v)(s) = \sqrt{c/2} \begin{cases} (p - v)(L - cs), & 0 < s < \ell, \\ (p + v)(c(s - \ell)), & \ell < s < \tau. \end{cases} \quad (4)$$

It sends the entire physical generator, including its domain, to

$$A_q w = w', \quad \text{Dom}(A_q) = \{w \in H^1(0, \tau) : w(\tau) = q w(0)\}. \quad (5)$$

Proof. The two changes of variables in (4), and $|p - v|^2 + |p + v|^2 = 2(|p|^2 + |v|^2)$, prove the isometry. For arbitrary w , the inverse is

$$p(x) = \frac{w(\ell + x/c) + w(\ell - x/c)}{\sqrt{2c}}, \quad v(x) = \frac{w(\ell + x/c) - w(\ell - x/c)}{\sqrt{2c}}, \quad (6)$$

$$u(x) = c^{-1} \int_0^x p(y) dy. \quad (7)$$

This lies in $\mathcal{H}$. For domain vectors the two midpoint traces agree because $v(0) = 0$. At the outer endpoints, $p(L) = -\eta v(L)$ gives $w(\tau) = q w(0)$, without division by q . Conversely those trace relations recover the physical boundaries and Sobolev regularity. Differentiation on each half proves $UG_\eta U^{-1} = A_q$. $\square$

In particular the dissipativity identity retains the physical constant:

$$\text{Re}\langle A_q w, w \rangle = \frac{1}{2}(q^2 - 1)|w(0)|^2 = -c\eta|v(L)|^2. \quad (8)$$

3 The complete evolution and spectrum

Theorem 2 (All times and all spectral points). *For $t \geq 0$, set $k = \lfloor (s + t)/\tau \rfloor$ for almost every $s \in (0, \tau)$. The strongly continuous semigroup and its exact norm are*

$$S_q(t)w(s) = q^k w(s + t - k\tau), \quad \|S_q(t)\| = |q|^{\lfloor t/\tau \rfloor}. \quad (9)$$

At $q = 0$, the coefficient means one for $k = 0$ and zero for $k \geq 1$. If $q \neq 0$, the entire spectrum is algebraically simple:

$$\lambda_n = \frac{\log |q| + i(\arg q + 2\pi n)}{\tau}, \quad n \in \mathbb{Z}. \quad (10)$$

The functions $\tau^{-1/2}e^{\lambda_n s}$ form a complete Riesz basis. If $q = 0$, the spectrum is empty and the sharp extinction time is τ .

Proof. Characteristics give (9). Floor addition proves the semigroup law; translation continuity on a dense set, followed by the contraction estimate, proves strong continuity. For $t = n\tau + a$ with $0 \leq a < \tau$, the two translated input intervals have disjoint supports and weights q^n, q^{n+1} . The first has positive length $\tau - a$. Integration proves the norm bound, and data supported on that first input interval attain it. This includes the exact endpoints $t = n\tau$.

Solving $zw - w' = f$ gives the kernel

$$R(z, A_q)(s, r) = e^{z(s-r)} \left[\frac{e^{z\tau}}{e^{z\tau} - q} - \mathbf{1}_{r < s} \right], \quad e^{z\tau} \neq q. \quad (11)$$

It is a bounded kernel mapping into (5); direct substitution verifies both inverse identities. At each excluded point the exponential is an eigenvector, so no residual or continuous spectral points are omitted. For $\alpha = (\log |q| + i \arg q)/\tau$, multiplication by $e^{\alpha s}$ conjugates the periodic derivative plus α to A_q . The Fourier basis then proves the Riesz and algebraic-simplicity statements. At $q = 0$, (11) exists everywhere and reduces to

$$R(z, A_0)f(s) = \int_s^\tau e^{z(s-r)} f(r) dr, \quad z \in \mathbb{C}. \quad (12)$$

Its locally uniform exponential series is operator-norm entire. Equation (9) is norm one for $t < \tau$ and zero for $t \geq \tau$, proving sharp extinction independently of spectral language. $\square$

The condition number of the specified multiplication similarity is $1/|q|$, not an asserted optimal condition number among all possible bases. For $q = -1$ the group is unitary. Compact resolvent and empty spectrum are compatible for an unbounded non-self-adjoint generator; the nonempty-spectrum theorem for bounded complex operators does not apply to A_0 .

η	q	$\ S_q(n\tau)\ , n \geq 1$	Imaginary spectral ladder
0	-1	1	$(2n + 1)\pi/\tau$
1/2	-1/3	3^{-n}	$(2n + 1)\pi/\tau$
1	0	0	No spectral points
2	1/3	3^{-n}	$2n\pi/\tau$

Table 1: Exact parameter controls. Equal decay norms do not determine the spectral ladder. The ladder index ranges over all integers, independently of the positive time-step index in the norm column.

4 Executable evidence and scope

The canonical ledger uses seven impedances, three round-trip times and seven time ratios. Four spatial ratios give 588 exact transport rows; seven boundary rows check the reflection sign and dissipation. A separate checker uses repeated boundary crossings instead of the producer’s floor formula. It checks 126 finite spectral rows by the exponential boundary equation, and 21 Green rows by direct kernel integration instead of the producer’s polynomial inhomogeneous solution. The stored decimal values have 60 significant digits from 100-digit arithmetic. They are not interval certificates. Canonical reproduction, two-directory replay and repaired-hash mutations test the artifact contract; they cannot establish the all-time and full-spectrum theorem by enumeration. Unknown fields and Boolean-integer substitutions are rejected by typed checks. The frozen evaluation is parsed strictly and locked before a release-write operation. Compiler logs, font checks and PDF hashes are separate engineering evidence, not mathematical novelty.

The conservative Route-A tuple is four failures followed by a formal operator hint; overall the candidate is rejected for the target route. The native generator supplies no rational-prime carrier, arithmetic primitive correspondence, target divisor, target functional equation or Hilbert–Pólya identification. No Euler factors, root number, automorphy or target local data are claimed. Route B remains disabled.

NO_BAD_EULER_OR_ROOT_NUMBER.

Literature and review boundaries. The author’s website PDF initially failed to download. Retrieval from the same author’s GitHub repository succeeded, and the relevant extracted text was read. The optional structural preflight was unavailable because its PDF-parser dependency was absent; no certified local-page-anchor claim is made. Equations (14)–(19), (25) and Theorem 1 of [1] support the ownership statement. The three-branch norm is established in the separate complete proof package; it is not attributed to a formula absent from the cited theorem. An internal team agent independently checked the complete analytic proof. The authoring agent supplied the implementation and manuscript. This is AI-assisted internal review, not external peer review or venue acceptance. Variable coefficients, internal damping and infinite impedance are outside the frozen theorem. They would require new domains and proofs.

5 Conclusion

Round zero: the physical domain and complete spectrum. The same exact unfolding determines all spectral points and all evolution times. At transparency the generator has no spectrum, yet the norm remains one until the sharp physical round-trip time.

References

- [1] T. A. Driscoll and L. N. Trefethen, Pseudospectra for the wave equation with an absorbing boundary, *Journal of Computational and Applied Mathematics* **69** (1996), 125–142. doi:10.1016/0377-0427(95)00021-6.